2 Transfer Function

LINEAR SYSTEM:

The system is termed as linear system if it obeys principle of superposition and principle of homogeneity.

It is basically a combination of two laws:

(a) Principle of superposition

(b) Principle of homogeneity

(a) Principle of superposition:

It states that the response produced by the simultaneous application of two distinct forcing function is the summation of the two individual responses.

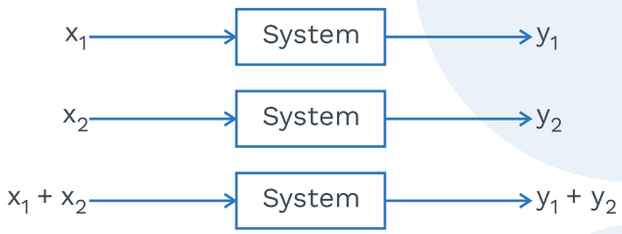


Figure: 2.1 Principle of Superposition:

(b) Principle of homogeneity:

It states that the output of a linear system is always directly proportional to the input.

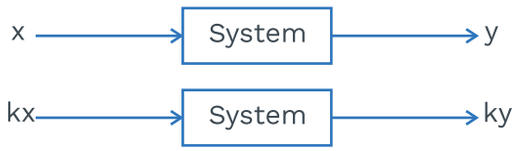


Figure: 2.2

Time Invariant System:

Consider a system in which, an input 'x(t)' is applied at 't' time and an output 'y(t)' is derived at the same time which is shown in figure 2.3.

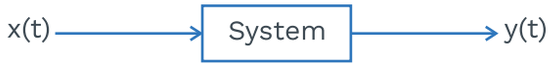


Figure: 2.3

Definitions

In control system, transfer functions are commonly used to characterize the input-output relationship of components or systems that can be described by a linear, time-invariant differential equations.

Definitions

The system is **time invariant system** if the relationship between input and output does not depend on time.

If we apply a delayed input by 'T' times i.e., x(t – T) then for the time invariant system output must be delayed by 'T' times as shown below.



Figure: 2.4

TRANSFER FUNCTION:

Transfer function of an LTI system is defined as the ratio of Laplace transform of output variable to Laplace transform of input variable, assuming all the initial conditions of system to be zero.



Figure: 2.5

$$T.F. = \frac{\mathcal{L}c(t)}{\mathcal{L}r(t)} = \frac{C(s)}{R(s)}$$

$$T.F. = \frac{C(s)}{R(s)}\Big|_{\text{Initial condition = 0}}$$

Poles and zeros of a transfer function:

The transfer function of a linear control system can be expressed in the form of a quotient of polynomials in the following form-

$$T(s) = \frac{a_0 s^m + a_1 s^{m-1} + a_2 s^{m-2} + \ldots\ldots\ldots\ldots a_m}{b_o s^n + b_1 s^{n-1} + b_2 s^{n-2} + \ldots\ldots\ldots\ldots b_n}$$

$$T(s) = \frac{a_0\left(s^m + \frac{a_1}{a_0} s^{m-1} + \ldots\ldots\ldots\ldots \frac{a_m}{a_0}\right)}{b_o\left(s^n + \frac{b_1}{b_o} s^{n-1} + \ldots\ldots\ldots\ldots \frac{b_n}{b_0}\right)}$$

The numerator and the denominator can be factored into m and n terms respectively, then the above expression for the transfer function can be expressed as-

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Any system that does not satisfy the superposition principle or its output is not directly proportional to its input is said to be non linear system.

Rack Your Brain

Why transfer function analysis is not applicable to non-linear systems?

$$T(s) = \frac{K(s-z_1)(s-z_2).....(s-z_m)}{(s-p_1)(s-p_2).....(s-p_n)}$$

Where, $K = \frac{a_o}{b_o}$

- K is called the gain factor of transfer function (T.F.)
- $z_1, z_2, ..., z_m$ is called the "Zeros" of transfer function.
- $p_1, p_2, ..., p_n$ (roots of denominator) is called the poles of the transfer function.
- In any physical realizable system p > z.
- In proper transfer function,
  Total number of poles > Total number of zeros
- If poles and zeros are not equal then we will assume that some poles and zeros lie at infinite to make the transfer function proper.

Type and Order of the System:

- 'Order' of System = Total number of poles of CLTF
- 'Type' of System = Total number of poles that lie at origin of OLTF.

The control loop configuration is basically of two types:

(a) Closed Loop Control System:

The closed loop configuration is characterized by the feedback signal. The changes in the output are measured through feedback and compared with input to achieve the control objective.

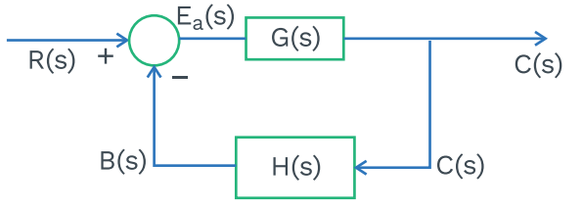


Figure: 2.6 Closed loop system

Forward path T.F. $= \frac{C(s)}{E_a(s)} = G(s)$

Rack Your Brain

Why transfer functions always defined under zero initial conditions?

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- At the 'zeros' of the T.F., the magnitude of the transfer function is zero.
- At the 'poles' of the T.F. the magnitude of the transfer function will be infinite.

Feedback path TF $= \frac{B(s)}{C(s)} = H(s)$

Closed loop T.F. $= \frac{C(s)}{R(s)}$

From figure 2.6,

$$E_a(s) = R(s) - B(s)$$

$$\frac{C(s)}{G(s)} = E_a(s)$$

$$\frac{C(s)}{G(s)} = R(s) - B(s)$$

And $B(s) = C(s) \cdot H(s)$

So,

$$\frac{C(s)}{G(s)} = R(s) - C(s)H(s)$$

$$\Rightarrow \quad C(s)[1 + G(s) \cdot H(s)] = G(s)R(s)$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

Hence, Closed Loop Transfer Function (CLTF) $= \frac{G(s)}{1 + G(s)H(s)}$

(b) Open loop control System:

In this configuration, feedback is not present.



Figure: 2.7

Open loop transfer function = G(s)

Example-1

What is the type and order of the system which is given below:

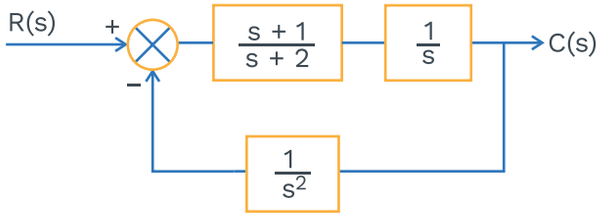


Figure: 2.8

Solution:

To find the type of the system, we have to make H(s) = 1

Now, first we will calculate the transfer function of the given system,

$$CLTF = \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{(s+1)}{(s+2)}\frac{1}{(s)}}{1 + \frac{(s+1)}{(s+2)s^3}}$$

$$CLTF = \frac{s^2(s+1)}{s^3(s+2) + s + 1} = \frac{\text{Numerator}}{\text{Denominator}}$$

Now, OLTF for (H(s) = 1)

$$OLTF = \frac{\text{Numerator}}{\text{Denominator} - \text{Numerator}}$$

$$OLTF = \frac{s^2(s+1)}{s^3(s+2) + s + 1 - s^2(s+1)} = \frac{s^2(s+1)}{s^4 + s^3 - s^2 + s + 1}$$

As we can see in the equation of OLTF, there is no pole. So, the type of the system = 0 and from CLTF the order of the system = 4.

Shortcut Method:

Valid only for unity feedback system

1. To find CLTF from OLTF

If $OLTF = \frac{\text{Numerator (N)}}{\text{Denominator (D)}}$

Then,

$$CLTF = \frac{\text{Numerator}}{\text{Numerator} + \text{Denominator}}$$

2. To find OLTF from CLTF

If $CLTF = \frac{\text{Numerator}}{\text{Denominator}}$

Then, $OLTF = \frac{\text{Numerator}}{\text{Denominator} - \text{Numerator}}$

Gray Matter Alert!!!

If the initial conditions are not zero in any system then we cannot calculate output from the transfer function i.e.

$$Y(s) \neq X(s).\ H(s)$$

- Transfer function can be written in two forms-
  (a) Pole – Zero form
  (b) Time constant form

(a) Pole – Zero form:

In this type of form we can easily find the total number of poles and zeros and also find out the location of zeros and location of poles.

$$\text{T.F.} = \frac{C(s)}{R(s)} = \frac{\text{Numerator}}{\text{Denominator}}$$

$$\text{T.F.} = \frac{K(s-s_a)(s-s_b).....(s-s_m)}{(s-s_1)(s-s_2).....(s-s_n)}$$

Where, Zeros : s_a, s_b, ..., s_m
Poles : s_1, s_2, ..., s_n

(b) Time constant form:

$$TF = \frac{C(s)}{R(s)} = \frac{K(1+T_{z1}s)(1+T_{z2}s)+.....}{(1+T_{P1}s)(1+T_{P2}s)+.....}$$

Where, T_{Z1}, T_{Z2}, ... T_{P1}, T_{P2} .....are time constants.

DC gain of transfer function:

Magnitude of transfer function at zero frequency gives the DC gain of transfer function.

Let a transfer function having 'p' number of poles at origin can be written as-

$$TF = \frac{K(s-s_a)(s-s_b)_____(s-s_m)}{s^p(s-s_1)(s-s_2)_____(s-s_n)}$$

The DC gain of TF can be find out by putting s = 0 in T.F.

At DC gain, $\omega = 0$

Put $s = j\omega = 0$ in transfer function, we get-

$$\therefore \quad \text{DC gain} = \lim_{S\to 0} \frac{K(s-s_a).....(s-s_m)}{s^p(s-s_1).....(s-s_n)}$$

Gray Matter Alert!!!

Every zero has the time constant and every pole has the real time constant.

Example-2

A pole-zero location at complex plane is shown in the figure below. If the DC gain of system is 2 then find out the transfer function?

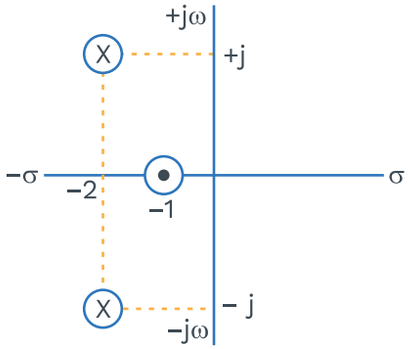


Figure: 2.9

Solution:

From the pole – zero diagram,

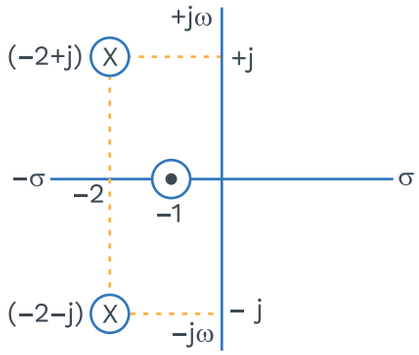


Figure: 2.10

Zero = –1

Pole = –2 + j, –2 –j

So transfer function in pole-zero form,

$$TF = \frac{K(s+1)}{(s+2-j)(s+2+j)}$$

$$TF = \frac{K(s+1)}{(s+2)^2+1}$$

Since, DC gain = 2

Put s = 0 in transfer function,

$$\text{DC gain} = 2 = \frac{K(0+1)}{(0+2)^2+1}$$

$$\frac{K}{5} = 2$$

K = 10

So, $TF = \frac{10(s+1)}{(s+2)^2+1}$

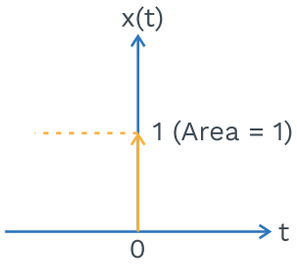
STANDARD SIGNALS:

1. Unit Impulse Signal:

$x(t) = \delta(t)$

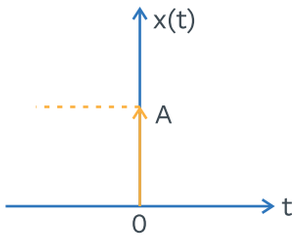
Where, $\delta(t) = \begin{cases} 0, & t \neq 0 \\ \infty, & t = 0 \end{cases}$

And $\int_{-\infty}^{\infty} \delta(t).dt = 1$



2. Impulse Signal:

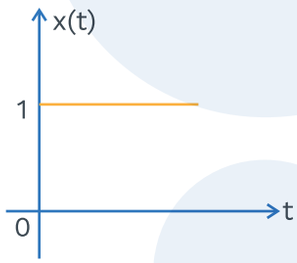
$x(t) = A\delta(t)$



Where, $x(t) = \begin{cases} A.\delta(t) = 0, & t \neq 0 \\ A.\delta(t) \to \infty, & t = 0 \end{cases}$

And $\int_{-\infty}^{\infty} A.\delta(t)\,dt = A$

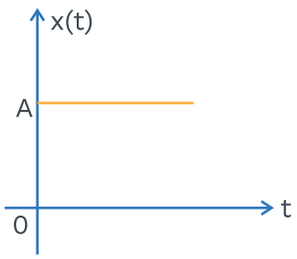
3. Unit Step Signal:



$x(t) = u(t)$

$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$

4. Step Signal:

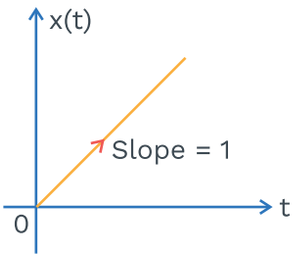


$x(t) = Au(t)$

$x(t) = \begin{cases} A, & t \geq 0 \\ 0, & t < 0 \end{cases}$

At $t = 0$, $Au(t) = 0.5$

5. Unit Ramp Signal:



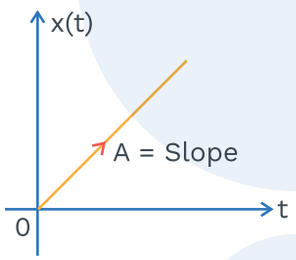
$$x(t) = tu(t)$$

$$x(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

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1. Impulse signals have constant area.
2. Step signals have constant amplitude.
3. Ramp signal have a constant slope.

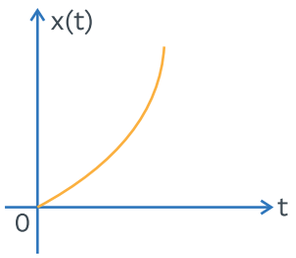
6. Ramp Signal:



$$x(t) = Atu(t)$$

$$x(t) = \begin{cases} At, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

7. Parabolic Signal:

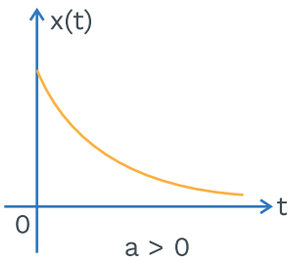


$$x(t) = \frac{At^2}{2}u(t)$$

$$x(t) = \begin{cases} \frac{At^2}{2}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

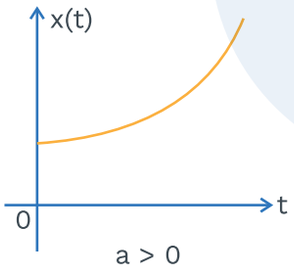
8. One sided exponential:

$x(t) = e^{-at}u(t)$



$x(t) = e^{at}u(t)$

↓



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$\frac{d}{dt}$ (Step Response) = Impulse Response

$\frac{d}{dt}$ (Ramp Response) = Step Response

$\frac{d}{dt}$ (Parabolic Response) = Ramp Response

Relation between impulse, unit and ramp signal:

$$\frac{d}{dt}(tu(t)) = u(t) \quad ; \quad \int_{-\infty}^{t} u(t).dt = tu(t)$$

or

$$\frac{du(t)}{dt} = \delta(t) \qquad \int_{-\infty}^{t} \delta(t).dt = u(t)$$

Transfer function and impulse response function:

Consider a linear time invariant system has the input r(t) and output c(t).



Figure: 2.11

Impulse response is defined as the output when the input is a unit impulse function $\delta(t)$.

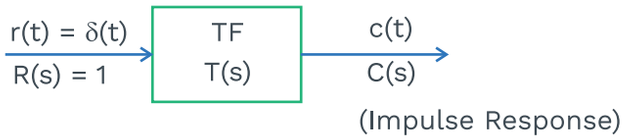


Figure: 2.12

$C(s) = R(s) \cdot T(s)$

$R(s) = 1$

$\therefore \quad C(s) = T(s)$

Impulse response = T(s)

Where T(s) is the transfer function of a system. Once the impulse response of a linear system is known, the output of a system c(t) with any input r(t), can be found by using transfer function.

$$T.F = T(s) = \text{Impulse response}$$

$$T(s) = C(s)$$

$$T(s) = L[c(t)]$$

Transfer function = LT [Impulse response]

$$\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 10y = 4x + 5\frac{dx(t)}{dt}$$

$$\left(s^2 + 7s + 10\right)Y(s) = (4 + 5s)X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{5s + 4}{s^2 + 7s + 10}$$

Impulse response = $\mathcal{L}^{-1}$ (Transfer function)

$$= \mathcal{L}^{-1}\frac{5s + 4}{(s + 2)(s + 5)}$$

$$= \mathcal{L}^{-1}\left[-\frac{2}{s + 2} + \frac{7}{s + 5}\right]$$

$$= -2e^{-2t}u(t) + 7e^{-5t}u(t)$$

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- Transfer function is the characteristic of the system. It does not depend upon input signal.
- Transfer function is defined only for a linear time – invariant system.

Previous Year's Questions

Let a causal LTI system be characterized by the following differential equation, with initial rest condition,

$$\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 10y(t) = 4x(t) + 5\frac{dx(t)}{dt}$$

Where, x(t) and y(t) are the input and output respectively. The impulse response of the system is (u(t) is the unit step function) **[GATE-2017]**

(1) $2e^{-2t}\,u(t) - 7e^{-5t}\,u(t)$

(2) $-2e^{-2t}\,u(t) + 7e^{-5t}\,u(t)$

(3) $7e^{-2t}\,u(t) + 2e^{-5t}\,u(t)$

(4) $-7e^{-2t}\,u(t) + 2e^{-5t}\,u(t)$

Solution: (2)

FEEDBACK AND NON-FEEDBACK SYSTEM:

Non feedback system:

A non-feedback open loop system which is represented by a block diagram as shown in figure 2.13, is activated by a single signal at the input (for single – input system)



Figure: 2.13 A non feedback (open – loop) system

- There is no provision which is provided within the system for the supervision of output. There is unavailability of any mechanism to compensate the system's behaviour for any lack of proper performance. The performance includes system components, loading, changing environment or neglection of exact values of process parameters.

Definitions

Feedback is a mean of automatic control and regulation. It can be achieved in many biological, physical and soft systems. For instance, the temperature of any human being can be automatically regulated by using a process named as feedback process.

Feedback system

- A feedback system with closed loop is represented by the block diagram in figure 2.14. This system drives by two or more signals, one of the signal is input signal while other is feedback signal which is derived from the output of the system. The feedback signal gives this system the capability to act as self correcting mechanism.

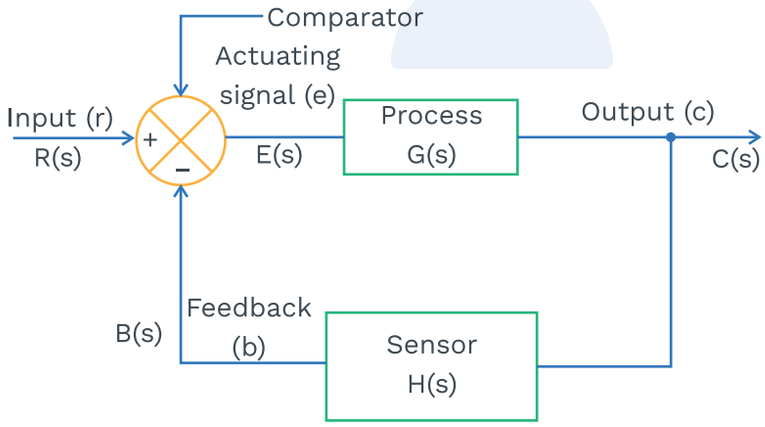


Figure 2.14: A feedback (closed loop) system

The sensor H(s) measures output signal c and this sensor produces a feedback signal b. The comparator tries to compare the input signal 'r' with feedback signal 'b' by generating the actuating signal 'e' . This actuating signal 'e' measures the difference between 'r' and 'b'.
The actuating signal is applied to the process G(s) so as to influence the output c in such a way that it leads to reduction in errors.

- The beneficial effects of feedback in a feedback system with high loop gain, are enumerated below:-
  1. The controlled variable precisely follows desired output.
  2. The effect due to external disturbance on controlled variable are largely reduced except the disturbances which are associated with the feedback sensors.
  3. Accurate control of the output is observed with the help of feedback in control loop even when controlled plant parameters or processes remains unknown or inaccurate.
  4. Feedback in control system vastly improves the speed of its response compared to the speed response capability of plants/components which comprises the system.

Effect of feedback on sensitivity:

The reduction in sensitivity of the system for parameter variation is one of the main purpose of using feedback in control system.

Let the variable in a control system which changes its value be α such as output and this change is considered due to parameter variation of element β such as gain or feedback then the control system sensitivity is expressed as:

$$\text{Sensitivity} = \frac{\%\,\text{Change in}\,\alpha}{\%\,\text{Change in}\,\beta}$$

In mathematical terms, sensitivity is written as-

$$S_{\beta}^{\alpha} = \frac{\%\,\text{Change in}\,\alpha}{\%\,\text{Change in}\,\beta}$$

$$= \frac{\frac{d\alpha}{\alpha}}{\frac{d\beta}{\beta}}$$

$$S_{\beta}^{\alpha} = \frac{\beta}{\alpha} \cdot \frac{d\alpha}{d\beta}$$

Definitions

Sensitivity – It is basically a measure of the effectiveness of the feedback which reduces the influence of random variations on system's performance.

Rack Your Brain

What are the necessary components of the feedback control system?

Sensitivity analysis of open loop control system:

For the open loop control system shown in figure. (2.15)



Figure: 2.15 Open loop system

The overall transfer function is

$$T(s) = \frac{C(s)}{R(s)} = G(s)$$

$$S_G^T = \frac{\%\,\text{Change in } T(s)}{\%\,\text{Change in } G(s)}$$

$$= \frac{\frac{\partial T(s)}{T(s)} \times 100}{\frac{\partial G(s)}{G(s)} \times 100}$$

$$S_{G(s)}^{T(s)} = \frac{G(s)}{T(s)} \cdot \frac{\partial T(s)}{\partial G(s)}$$

In open loop system T(s) = G(s)
Differentiating w.r.t. G(s)

$$\frac{\partial T(s)}{\partial G(s)} = 1$$

So, $$S_{G(s)}^{T(s)} = \frac{G(s)}{G(s)} \times 1$$

$$S_{G(s)}^{T(s)} = 1$$

The sensitivity of the overall transfer function with respect to the forward path transfer function in the case of closed loop control system is reduced by a factor [1 + G(s) H(s)] as compared to open loop system.

Gray Matter Alert!!!

- $S_G^T = 1 = 100\%$
- Percentage change of T(s) = Percentage change of G(s)
- $S_G^T = \frac{1}{1+GH}$
- Percentage change of, $T(s) = \frac{1}{1+GH} \times$ percentage change of G(s)

Sensitivity analysis of closed loop control system with respect to forward path-

For the closed loop control system shown in Figure (2.16) the overall transfer function is,



Figure: 2.16 Closed loop control system

$$T(s) = \frac{G(s)}{1 + G(s).H(s)} \quad \text{..... (I)}$$

Differentiating w.r.t. G(s)

$$\frac{\partial T(s)}{\partial G(s)} = \frac{\left[1+G(s)H(s)\right]-\left[G(s)H(s)\right]}{\left[1+G(s)H(s)\right]^2}$$

$$\frac{\partial T(s)}{\partial G(s)} = \frac{1}{\left[1+G(s)H(s)\right]^2} \quad \text{..... (II)}$$

$$S_{G(s)}^{T(s)} = \frac{G(s)}{T(s)} \cdot \frac{\partial T(s)}{\partial G(s)}$$

$$S_{G(s)}^{T(s)} = \frac{G(s)}{\frac{G(s)}{1+G(s)H(s)}} \cdot \frac{1}{\left[1+G(s)H(s)\right]^2}$$

$$S_{G(s)}^{T(s)} = \frac{1}{1+G(s)H(s)}$$

Sensitivity of overall transfer function T(s) with respect to feedback path transfer function.

- The sensitivity function for the overall transfer function T(s) with respect to variations in the feedback path transfer function H(s) is written as,

$$S_H^T = \frac{\frac{\partial T(s)}{T(s)}}{\frac{\partial H(s)}{H(s)}}$$

$$S_H^T = \frac{H(s)}{T(s)} \cdot \frac{\partial T(s)}{\partial H(s)} \quad \text{..... (I)}$$

The overall transfer function T(s) for the closed loop control system is,

$$T(s) = \frac{G(s)}{1+G(s).H(s)} \quad \text{..... (II)}$$

Differentiating w.r.t. H(s)

$$\frac{\partial T(s)}{\partial H(s)} = \frac{-\left[G(s)\right]^2}{\left[1+G(s).H(s)\right]^2} \quad \text{..... (III)}$$

Gray Matter Alert!!!

- A closed loop control system is more sensitive to variations in feedback path parameters than the variation in forward path parameters therefore, the specifications of feedback elements in a closed-loop control system should be more rigid as compare to that of forward path.
- In open loop system, the sensitivity due to feedback element is nill (zero).

Previous Year's Questions

As shown in the figure, a negative feedback system has on amplifier of gain 100 with ± 10% tolerance in the forward path and on attenuator of value 9/100 in the feedback path. The overall system gain in the feedback path. The overall system gain in approximately. **[GATE-2010]**



(1) 10 ± 1 % (2) 10 ± 2 %
(3) 10 ± 5 % (4) 10 ± 10 %

Solution: (1)

Substituting equation (II) and (III) in equation (I)
The sensitivity of T(s) w.r.t. H(s) for a closed loop control system is obtained as-

$$S_H^T = \frac{-G(s)\,H(s)}{1+G(s)\,H(s)}$$

Effect of feedback on bandwidth:
It increases the bandwidth of system. Consider the elementary single loop feedback system as shown in figure (2.17).

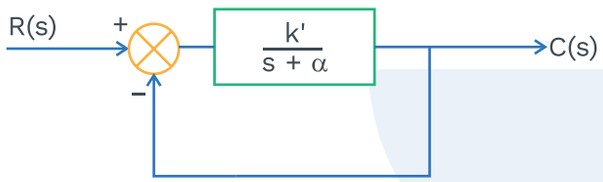


Figure:2.17 A simple feedback system

Definitions

Bandwidth- Bandwidth is the range of frequencies at which the gain of system is more than $\frac{1}{\sqrt{2}}$ times of its DC gain.

The open loop and closed loop transfer function can be expressed in the frequency domain (s = jω) as –

Open loop $G(s) = \frac{k}{1+\tau s}$; $\quad k' = \frac{k}{\alpha}, \tau = \frac{1}{\alpha}$

$$G(j\omega) = \frac{k}{1+j\omega\tau}$$

Closed loop T(s) = G/(1+GH)

$$T(s) = \frac{k/(1+k)}{1+j\omega\tau_c} \;;\quad \tau_c = \frac{\tau}{(1+k)}$$

$$T(j\omega) = \frac{k/(1+k)}{1+j\omega\tau_c}$$

Their bandwidth is determined as below –

$$(\omega_b\tau)^2 + 1 = \left(\sqrt{2}\right)^2$$

$$\omega_b(OL) = \frac{1}{\tau}$$

$$(\omega_b\tau_c)^2 + 1 = \left(\sqrt{2}\right)^2$$

$$\omega_b(CL) = \frac{1}{\tau_c}$$

Gray Matter Alert!!!

Closed loop system has a bandwidth (1 + k) times the bandwidth of the open loop system this implies increased speed to response because speed is proportional to bandwidth.

Where frequency ω_b is the bandwidth of the system, at which gain drops to $\frac{1}{\sqrt{2}}$ of its DC value.

$$\frac{\omega_b(CL)}{\omega_b(OL)} = \frac{\tau}{\tau_c} = (1+k)$$

Effect of feedback on stability:

The block diagram of the negative feedback control system is shown in the figure below.

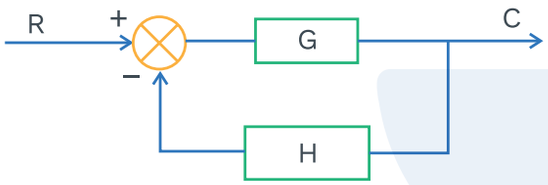


Figure: 2.18

Transfer function of negative feedback control system is,

$$T = \frac{G}{1+GH} \quad \ldots\ldots (I)$$

For the above expression, if GH = -1 then the system's output is infinite for any finite input and it is termed as unstable system. In general, GH=-1 is the only condition which leads to instability.

Consider the feedback system in figure (2.18) is unstable because value of GH = -1

If another feedback loop is introduced through a negative feedback gain of F which is shown in figure (2.19). The relation between input and output of the overall system is given as-

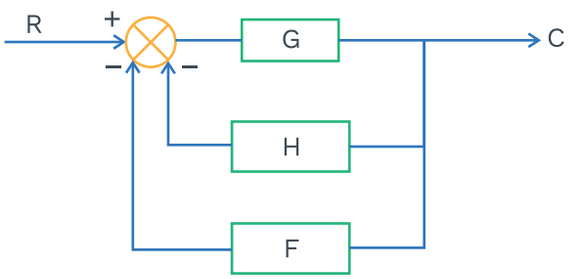


Figure: 2.19 Feedback system with two feedback loops

Definitions

Stability is defined as the system's ability to follow the input command which is generally helpful. If the output of the system is under control then it is termed as stable system otherwise it is termed as unstable system.

Rack Your Brain

What is the significance of bandwidth?

$$T = \frac{C}{R} = \frac{G}{1 + GH + GF}$$

It is possible that although G and H are such that the inner loop feedback system is unstable as GH = –1, the overall system may be stable by proper selection of the outer loop feedback gain F in practice. GH represents a function of frequency, and the stability condition of the closed loop system depends only on the phase and magnitude of GH.

Gray Matter Alert!!!

- The feedback may improve stability or may be harmful to the stability if it is applied in an improper way.
- It can be stated that feedback can cause a stable system to become unstable. Therefore, feedback can be stated as a double-edged sword, if it is inefficiently used, it becomes harmful.

Effect of feedback on overall gain:

- The gain G of a non-feedback system is affected by a feedback with a factor of 1 + GH.
- The overall gain of negative feedback closed loop control system is given as,

$$\text{Gain} = \frac{G}{1 + GH}$$

So, the overall gain can be decreased or increased which depends upon the value of 1+GH.

- If the value of 1 + GH is less than 1, then the overall gain increases. For this case 'GH' value is negative as the gain of the feedback path is negative.
- If the value of 1 + GH is greater than 1, then the overall gain decreases. For this case 'GH' value is positive as the gain of feedback path is positive.

Feedback effect on external disturbance or noise:

All physical system experiences some types of unwanted signals or noise during operation. External disturbances, like wind gusts acting on an antenna, are also quite frequent in control system.

The effect of feedback on noise and disturbance largely depends on the regions where these unwanted signals occur in the system. In many situations, feedback helps in reducing the effect of noise and disturbance on system performance. Refer to the system which is shown in figure (2.20), where R denotes the command.

Gray Matter Alert!!!

Practically, in case of a control system, G and H represents the functions of frequency where 1 + GH has magnitude may be greater than 1 in one frequency range whereas lesser than 1 in another frequency range. Hence, feedback can increase the gain of the system in one frequency range while decrease it in another frequency range.

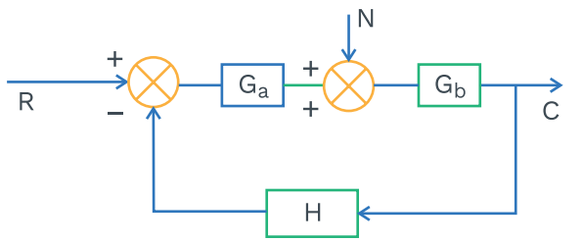


Figure: 2.20 Feedback system with noise signal

Gray Matter Alert!!!

To reduce the effect of disturbance, the magnitude of G_a should be high and magnitude of G_b should be low.

Signal N is the noise signal. In the absence of feedback, i.e. H = 0, the output C due to N acting alone is,

$$C = G_b\ N \qquad \text{..... (I)}$$

In the presence of feedback, output of the system due to N which is given as -

$$C = \frac{G_b N}{1 + G_a G_b H} \qquad \text{..... (II)}$$

On comparison of equation (I) with equation (II), the noise component in the output of equation (II) is reduced by the factor of $1 + G_a G_b H$, if the latter is greater than unity and the system remains stable.

Effect of feedback on transient response of system:

It improves the transient response of system by increasing the speed of system.

Let us consider an open loop system shown in figure (2.21).



Figure: 2.21

Rack Your Brain

What are the disadvantages of a negative feedback in a control system?

OLTF of the system,

$$T(s) = \frac{C(s)}{R(s)} = \frac{k}{s + \alpha}$$

For impulse input R(s) = 1, output for non-feedback system.

$$C(s) = \frac{k}{s + \alpha}$$

By taking inverse Laplace transform

$$c(t) = k.\ e^{-\alpha t}$$

If we convert the open loop system into closed loop system by introducing unity feedback then the overall transfer function for given figure 2.22 is,

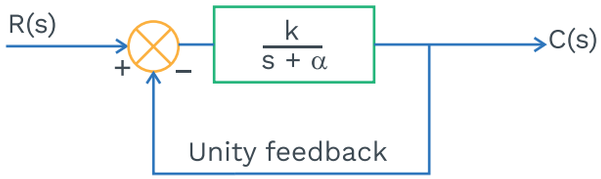


Figure: 2.22

$$\frac{C(s)}{R(s)} = \frac{k}{(s+\alpha+k)}$$

For impulse input R(s) = 1

$$C(s) = \frac{k}{(s+\alpha+k)}$$

By taking inverse LT

$$c(t) = k.\ e^{-(\alpha+k).t}$$

For positive value of k, the effect of the feedback is to shift the pole negatively to s = – (α + k), and so the time constant reduces. As k increases transient response decays more quickly.

Positive feedback system:

It is also called as regenerative feedback. In regenerative feedback, the output in feedback with positive sign as shown in figure (2.23).

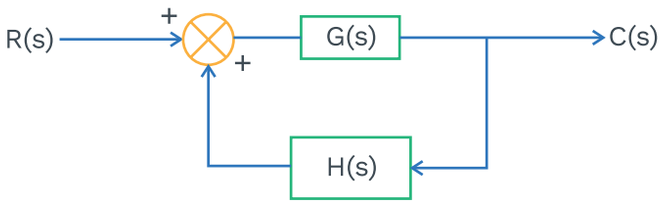


Figure: 2.23 Regenerative feedback control system

In this case, the transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1-G(s)H(s)}$$

Gray Matter Alert!!!

- The regenerative feedback is sometimes used for increasing the loop gain of feedback systems.
- It increases the gain of system by $\frac{1}{1-GH}$ factor.

Rack Your Brain

Why we generally do not use positive feedback in control system?

Previous Year's Questions

Feedback control systems are

(1) Insensitive to both forward and feedback path parameter changes.

(2) Less sensitive to feedback path parameter changes than to forward path parameter changes.

(3) Less sensitive to forward path parameter changes than to feedback path parameter changes.

(4) Equally sensitive to forward and feedback path parameter changes. **[GATE-2000]**

Solution: (3)

PRACTICE QUESTIONS

Q1 A control system with certain excitation is governed by the following mathematical equation

$$\frac{d^2x}{dt^2} + \frac{1}{2}\frac{dx}{dt} + \frac{1}{18} = 10 + 5e^{-4t} + 2e^{-5t}$$

The natural time constants of the response of the system are

(a) 2s and 5s
(b) 3s and 6s
(c) 4s and 5s
(d) 1/3s and 1/6s

Sol. Correct answer is (b)

Q2 A second order under damped system has a unit impulse response which starts from rest is given by $c(t) = 12.5e^{-6t} \sin 8t, t \geq 0$. What will be the steady state value of the unit step response for the given system?

(a) 0 (b) 0.25 (c) 0.5 (d) 1.0

Sol. Correct answer is (d)

Q3 The h(t) response of a LTI system to an impulse $\delta(t)$ for initially relaxed condition is $h(t) = e^{-t} + e^{-2t}$. What will be the response of the system for a unit step input u(t)?

(a) $u(t) + e^{-t} + e^{-2t}$
(b) $(e^{-t} + e^{-2t})\, u(t)$
(c) $(1.5 - e^{-t} - 0.5\, e^{-2t})\, u(t)$
(d) $e^{-t}\delta(t) + e^{-2t}u(t)$

Sol. Correct answer is (c)

Q4 For a system having transfer function, $G(s) = \frac{-s+1}{s+1}$, a unit step input is applied at time t = 0. The value of the response of the system at t = 1.5 sec (rounded off to three decimal places) is .....

Sol. 0.55-0.57

Q5 For the system governed by the set of equations:

$$\frac{dx_1}{dt} = 2x_1 + x_2 + u$$

$$\frac{dx_2}{dt} = -2x_1 + u$$

$$y = 3x_1$$

the transfer function $\frac{Y(s)}{U(s)}$ is given by

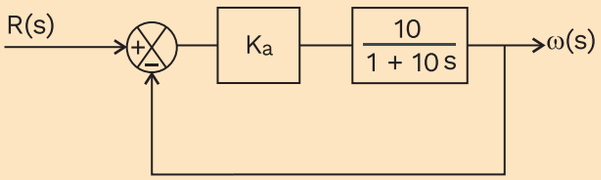
(a) $\frac{3(s+1)}{(s^2-2s+2)}$ (b) $\frac{3(2s+1)}{(s^2-2s+1)}$

(c) $\frac{(s+1)}{(s^2-2s+1)}$ (d) $\frac{3(2s+1)}{(s^2-2s+2)}$

Sol. Correct answer is (a)

Q6 The open-loop transfer function of a dc motor is given as $\frac{\omega(s)}{R(s)} = \frac{10}{1+10s}$.

A feedback is connected to the system as shown below. What will be the approximate value of K_a which reduces the time constant of the closed loop system by 1/100th times as compared to the open loop system.



(a) 1 (b) 5 (c) 10 (d) 100

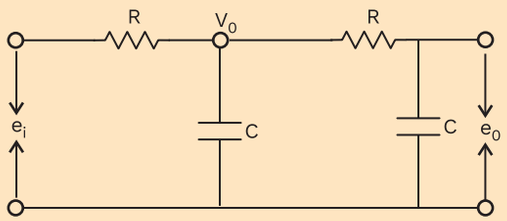
Sol. Correct answer is (c)

Q7 A unity feedback control system has a unit impulse response which is given by $c(t) = -te^{-t} + 2e^{-t}$, $t \geq 0$. Determine the open loop transfer function for the given system.

(a) $\frac{s+1}{(s+2)^2}$ (b) $\frac{2s+1}{s^2}$ (c) $\frac{s+1}{(s+1)^2}$ (d) $\frac{s+1}{s^2}$

Sol. Correct answer is (b)

Q8 The transfer function of the network shown below is



(a) $\frac{1}{s^2T^2 + 2sT + 1}$ (b) $\frac{1}{s^2T^2 + 3sT + 1}$ (c) $\frac{1}{s^2T^2 + sT + 1}$ (d) $\frac{1}{s^2T^2 + 1}$

Sol. Correct answer is (b)

Q9 Consider an LTI system initially at rest which is subjected to a unit step input. This provides a response of $y(t) = te^{-t}$, $t > 0$. What will be the transfer function of the given system?

(a) $\frac{1}{(s+1)^2}$ (b) $\frac{1}{s(s+1)^2}$ (c) $\frac{s}{(s+1)^2}$ (d) $\frac{1}{s(s+1)}$

Sol. Correct answer is (c)

Q10 The transfer function of a low pass RC network is

(a) $RCs(1+RCs)$ (b) $\frac{RC}{(1+RCs)}$ (c) $\frac{1}{(1+RCs)}$ (d) $\frac{s}{1+RCs}$

Sol. Correct answer is (c)